

Madhura Khandkar

717-802-1274 | madhura.k648@gmail.com | linkedin.com/in/madhurakhandkar | github.com/madhurakhandkar | madhurakhandkar.github.io

EDUCATION

The Pennsylvania State University

College of Engineering, Bachelor of Science in Computer Science

University Park, PA

Graduation: December 2026

Relevant Courses: Data Structures and Algorithms, Object Oriented Programming, Systems Programming, Computer Vision

SKILLS

Languages & Frameworks: Python, Java, JavaScript, PHP, Ruby, SQL, C++, React, Laravel, FastAPI, Ruby on Rails

Tools & Libraries: Git, Node.js, AWS Bedrock, GCP, MySQL, PostgreSQL, NumPy, Pandas, SciPy, Matplotlib

WORK EXPERIENCE

Mile6

Philadelphia, PA

Software Engineering Intern

May 2026 – August 2026

- Built an automated B2B lead generation tool in Laravel delivering PDF reports and follow-up email priority from a 30+ question client assessment
- Resolved a week-long production outage misreported as an email issue by tracing 3 independent root causes across the PDF generation pipeline
- Fixed an inventory bug affecting 1,000+ products across English and French translations by rewriting frontend display logic and correcting translation mapping

The Pennsylvania State University

University Park, PA

Software Engineering Intern

October 2025 – May 2026

- Consolidated data processing across distributed student and faculty systems by developing a Ruby on Rails API handling 50,000+ educational records
- Reduced manual processing by 60% by automating institutional data imports into Activity Insight with Ruby and XML export logic
- Identified and resolved 20+ bugs before release by leading PR reviews for the API and XML pipeline
- Shipped features using test-driven development in an Agile team across coding, QA, and deployment

The Pennsylvania State University

Middletown, PA

Undergraduate Researcher

May 2024 – December 2024

- Improved biomechanical movement accuracy by 25% by engineering a motorized 3D printed throat model
- Simulated anatomically accurate muscle contractions by programming Python controlled servo systems
- Designed CAD anatomical models and optimized motor placement for simulation purposes

PROJECTS

DigiLib

April 2026 – Present

Python, AWS Bedrock, React, LLaMA 3

- Developed a RAG-powered academic research tool that retrieves and ranks credible sources using conversational AI
- Integrated live OpenAlex search to query academic indexes from user prompts, retrieving only abstracts to avoid copyrighted material
- Won 2nd place and \$2,000 in the Bardusch Family IdeaMakers Challenge for the tool's sustainable business model

AI Voice Copilot

March 2026 – April 2026

Python, JavaScript, React, FastAPI, AWS Bedrock

- Created an AI voice copilot providing guidance to learn services on web platforms
- Implemented AWS Bedrock with Anthropic models to deliver recommendations at 800ms latency
- Captured user interactions across 3+ sites, streaming 30 clicks per session to FastAPI for validation

Creditwise

May 2025 – July 2025

React, Firebase, Figma

- Generated a web application enabling transfer students to verify credit transfer eligibility across institutions
- Reduced course search time by 30% by centralizing credit transfer data into a single student-facing platform
- Supported 30+ course equivalencies across 3 universities by establishing real-time data storage and evaluation logic

LEADERSHIP & INVOLVEMENT

AWS Builder Club

University Park, PA

Outreach Director

September 2025 – Present

- Expanded event participation from 20 to 100+ students by partnering with 20+ technical student organizations
- Coordinate technical outreach by organizing 3 AWS-focused workshops and speaker sessions
- Connect club members with industry professionals, increasing student exposure to new AWS tools